UID:

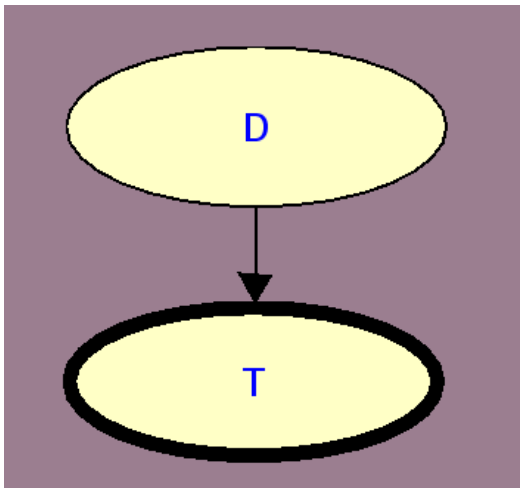
1.

P(D) = 0.001
P(~D) = 0.999
Pr(T|~D) = 0.02
Pr(~T|D) = 0.05

From given info:

Pr(~T|~D) = 1 - 0.02 = 0.98
Pr(T|D) = 1 - 0.05 = 0.95

a.



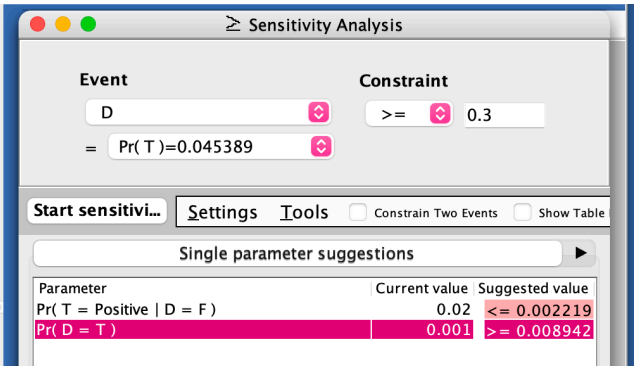
For D:

T	0.001
F	0.999

For T:

D	T	F
Positive	0.95	0.02
Negative	0.05	0.98

b.



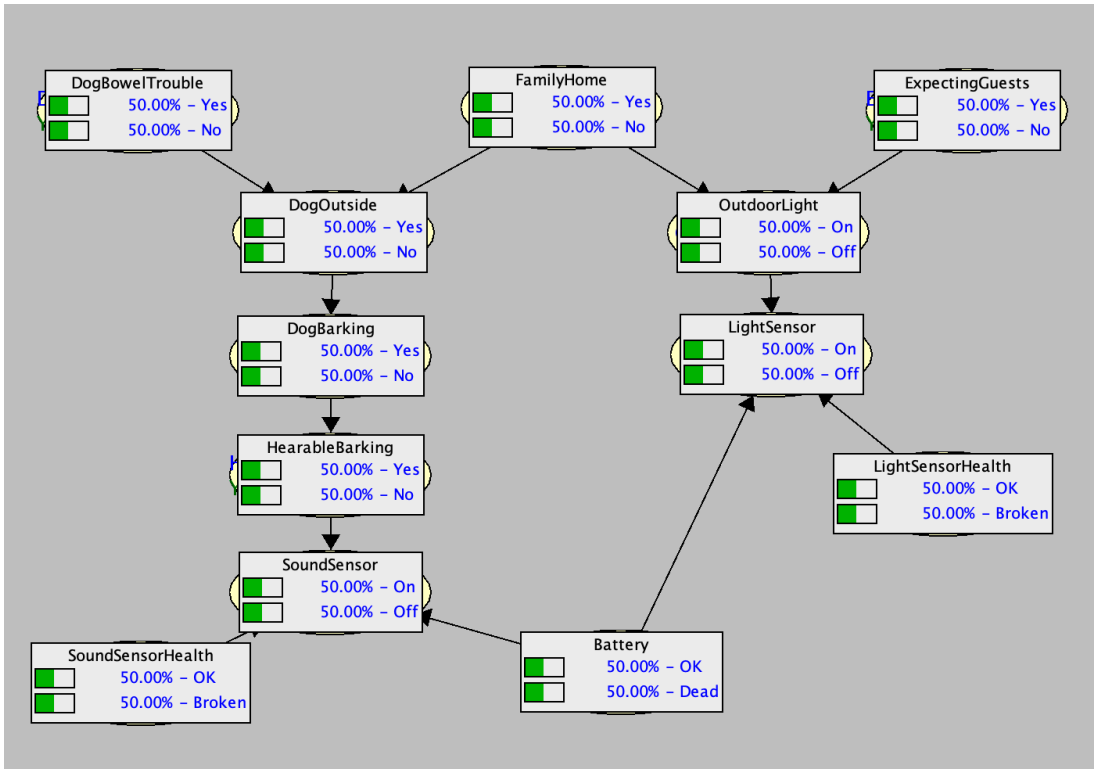
To constrain the Pr(D|T) to be greater than or equal to 0.3, we have to constrain Pr(T = Positive | D = F) (false positive rate) to <= 0.002219. For the prior probability of having the disease, Pr(D = T), >= 0.008942. The false negative probability does not appear since changing only that will probably not result in satisfying the constraint.

2.

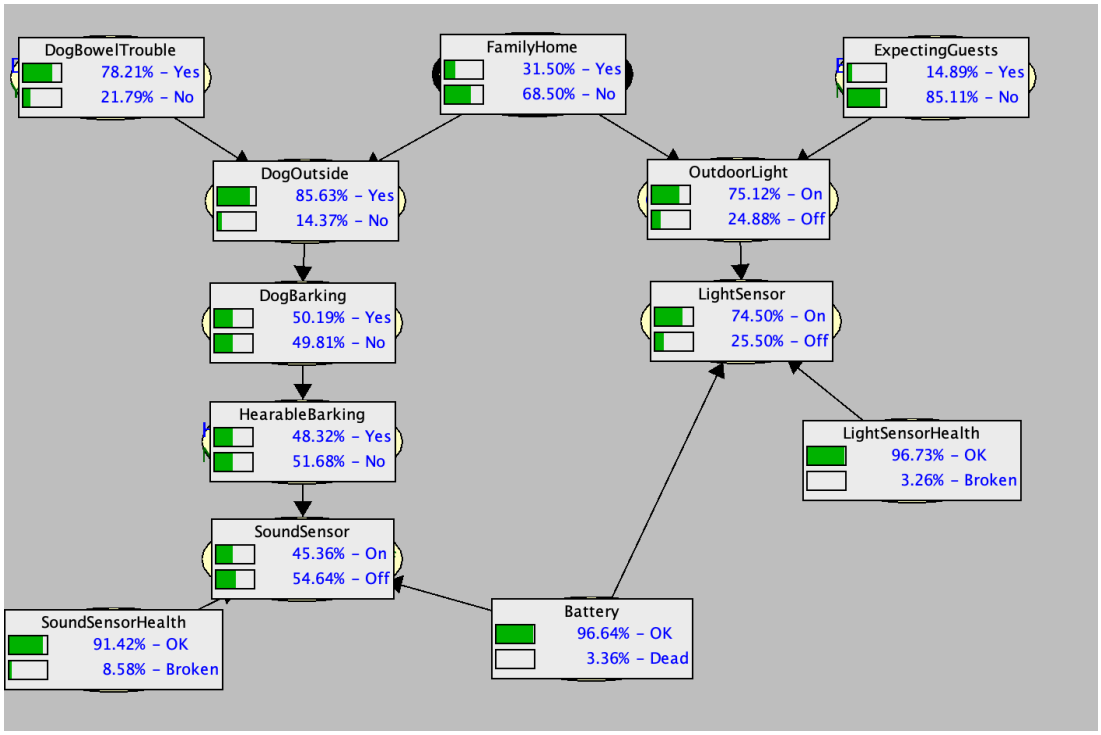
a.

Variables	**Values**
ExpectingGuests	Yes/No
FamilyHome	Yes/No
SoundSensor	On/Off
LightSensor	On/Off
HearableBarking	Yes/No
Battery	OK/Dead
SoundSensorHealth	OK/Broken
LightSensorHealth	OK/Broken
DogBarking	Yes/No
DogOutside	Yes/No
OutdoorLight	On/Off
DogBowelTrouble	Yes/No

Initial network with uniform parameters:

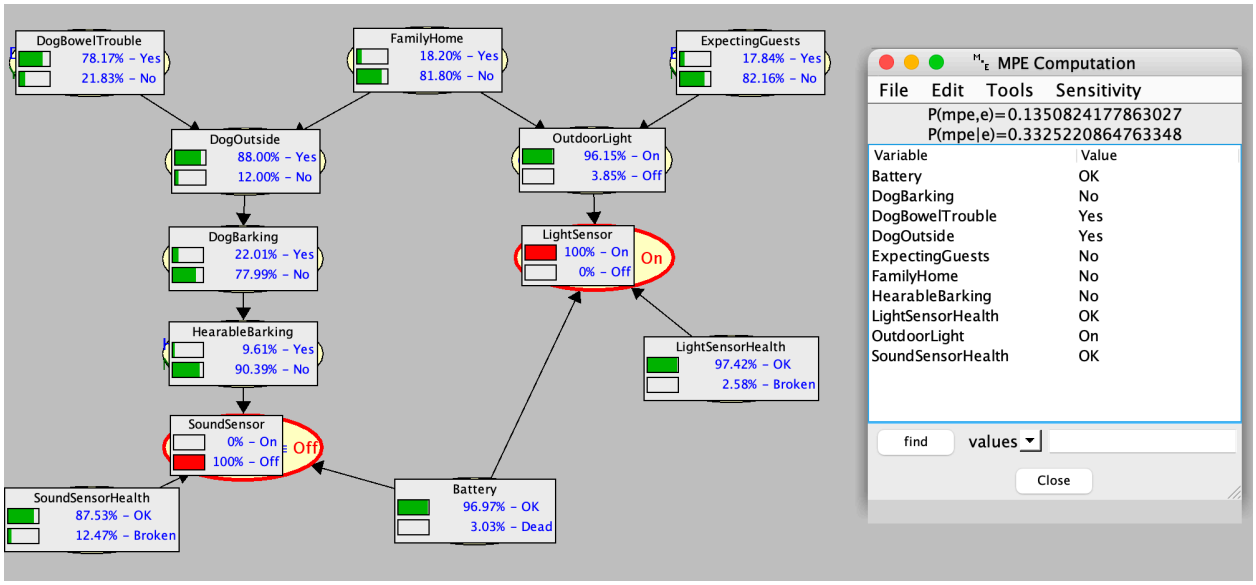


After running EM algorithm to learn network CPTs:



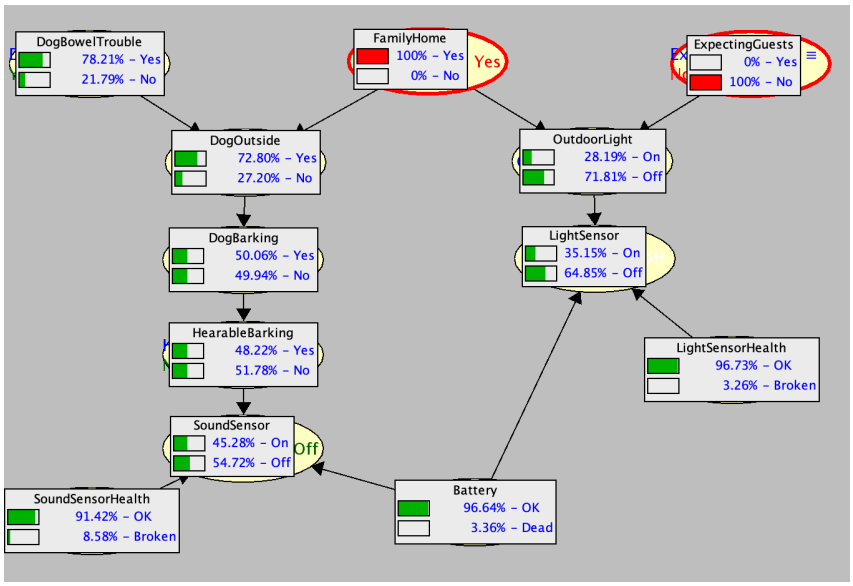
I interpreted HearableBarking as their dog barking results in a change in HearableBarking.

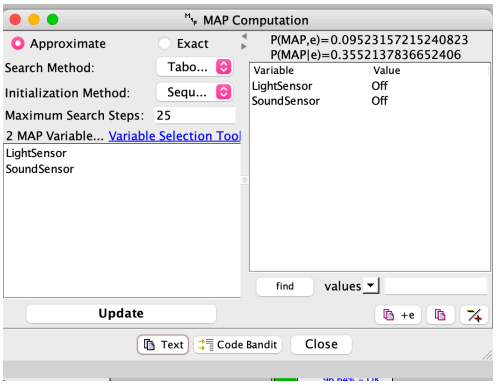
b.



I used MPE since we are looking at the most likely instantiation of *all* variables given that Sambot has sensed the lights to be on but has sensed no bark. I set LightSensor to On and SoundSensor to Off for this. The instantiations are shown above.

c.





I used MAP computation since we are trying to find the most likely instantiation of specific variables (the sensors) given FamilyHome = Yes and Expecting Guests = No. I set these values and then ran MAP for the LightSensor and SoundSensor. Both values are likely Off.

d.

Paths from SoundSensor to LightSensor:

- SoundSensor ← Battery → LightSensor

- SoundSensor ← HearableBarking ← DogBarking ← DogOutside ← FamilyHome → OutdoorLight → LightSensor

Z = {Battery, FamilyHome}

The sensors are d-separated by these values in Z since all paths would be blocked. The first is divergent/a fork, so if Battery is given that path is blocked. The second is a chain that includes a fork in the middle at FamilyHome, so adding FamilyHome to Z would also block that path. Since this results in d-separation, SoundSensor and LightSensor are independent given Z.

e. The network I created is a multiply connected network, since there are multiple paths between at least one pair of nodes. If we look back at part d, there are two paths between SoundSensor and LightSensor. It cannot be a tree since nodes have multiple parents, like DogOutside has parents DogBowelTrouble and FamilyHome. It cannot be a polytree because there is more than one undirected path between some nodes, like LightSensor and SoundSensor (discussed in d).